

# Solution to Homework 3, Question 4 (Poisoning Attack)

## Question Summary

In a targeted poisoning attack, we aim to misclassify a target test point by modifying training data points to influence the model's decision boundary. The attacker can perturb certain data points to achieve this misclassification.

## Given Data

- Model:  $y = aX + b$  with parameters  $a = [0.3, 0.1, 0.2]$  and  $b = 0$ .
- Threshold:  $y \geq 1$  is classified as 1; otherwise, it is 0.
- Training Points:
  - $x_1 = [1, 1, 3], y_1 = 1$
  - $x_2 = [1, 2, 2], y_2 = 0$
  - $x_3 = [2, 1, 0], y_3 = 1$
- Target Test Point:  $x_4 = [3, 0.5, 0.4], y_4 = 1$

## Part 1 Solution

**Goal:** Select one data point to perturb by adding  $\delta = 1$  to the first element of the  $x$ -vector and determine the new  $y$ -label for this perturbed point.

### Steps

1. Calculate the initial outputs for each  $x_i$  using the linear model  $y = aX + b$ :

$$y_1 = 0.3 \times 1 + 0.1 \times 1 + 0.2 \times 3 = 1.0$$

$$y_2 = 0.3 \times 1 + 0.1 \times 2 + 0.2 \times 2 = 0.9$$

$$y_3 = 0.3 \times 2 + 0.1 \times 1 + 0.2 \times 0 = 0.7$$

2. Apply the perturbation  $\delta = 1$  to the first element of each  $x_i$  and calculate the new  $y$ -value for each perturbed  $x_i$ :

- Perturb  $x_1$  to  $[2, 1, 3]$ :

$$y'_1 = 0.3 \times 2 + 0.1 \times 1 + 0.2 \times 3 = 1.3$$

- Perturb  $x_2$  to  $[2, 2, 2]$ :

$$y'_2 = 0.3 \times 2 + 0.1 \times 2 + 0.2 \times 2 = 1.2$$

- Perturb  $x_3$  to  $[3, 1, 0]$ :

$$y'_3 = 0.3 \times 3 + 0.1 \times 1 + 0.2 \times 0 = 1.0$$

3. **Selection:** Perturbing  $x_2$  results in the largest increase in output (from 0.9 to 1.2), making it the best choice for influencing the decision boundary.

4. The new  $y$ -label for the perturbed  $x_2$  is 1 since  $y'_2 = 1.2 \geq 1$ .

**Answer for Part 1:** Select  $x_2$  and the new  $y$ -label is 1.

## Part 2 Solution

**Goal:** Modify  $x_3$  (i.e.,  $V_p = \{x_3\}$ ) within a perturbation bound  $\epsilon = 0.5$  on each feature to achieve a clean-label attack.

### Steps

1. **Perturbation Constraint:** Each element in  $x_3$  can be adjusted by  $\pm 0.5$  to remain within the bound  $\|\delta\|_\infty \leq 0.5$ .
2. Apply the perturbation by increasing each element of  $x_3$  by 0.5 to get  $[2.5, 1.5, 0.5]$ .
3. Calculate the new  $y_3$  for the modified  $x_3$ :

$$y'_3 = 0.3 \times 2.5 + 0.1 \times 1.5 + 0.2 \times 0.5 = 1.0$$

**Answer for Part 2:** Modify  $x_3$  to  $[2.5, 1.5, 0.5]$ .

## Part 3 Solution

**Goal:** Determine the reasoning behind dropping data points in small clusters in the gradient space in the EPIC algorithm.

### Analysis

- **Option D** ("Effective poisons are closer to the target in embedding space") is correct. In the EPIC algorithm, effective poisons are expected to cluster near the target in embedding/gradient space to maximize their influence. Dropping isolated points in gradient space removes less impactful points.

**Answer for Part 3:** Option *D* – Effective poisons are closer to the target in embedding space.